

Response to the Editor and Reviewers

Submission ID: 20e66410-c472-42dc-ab99-957c24a43f96

Original title: Auditing observability bias in InSAR-derived subsidence exposure estimates with a reproducible geospatial workflow

Revised title: Quantifying how spatial allocation and sampling alter population-weighted InSAR support summaries

We thank the Editor and Reviewers for helping us sharpen the contribution and its evidential scope. The revised study makes a positive, testable claim: for a fixed population model, geographic domain and InSAR support field, spatial allocation and aggregation can materially change population-weighted support summaries. We quantify the size and direction of this effect, identify conditions that produce or reverse it, and test sensitivity to sampling and population models. We have removed the stronger interpretation of the original 3.77-million figure as a verified count of people exposed to subsidence in unobserved locations; this correction narrows the interpretation while preserving the supported methodological and empirical findings.

In addition to the requested reframing and point-by-point revisions, we conducted several further analyses to test where the conclusions hold and where they do not. These include a direct comparison of mean input-pair support with the final LiCSBAS quality mask and finite-domain bounds for unsupported pixels; controlled masking experiments that compare within-unit prevalence, centre sampling and a published unit-median-rate rule across missingness patterns and grid origins; crossed population-model and pair-sampling tests, plus scale/origin and fine-resolution allocation sensitivities; and Fresno benchmarks against Census blocks, CA-POP and ACS survey estimates. These additions test the robustness, transferability and limits of the revised claims. They do not provide physical validation in genuinely unsupported areas, and we state those limits alongside each result.

Location convention: main-manuscript line ranges refer to the supplied `Manuscript_Numbered.pdf`; page numbers refer to the clean manuscript and Supplementary Information PDFs. Each response lists the relevant final-manuscript lines and Supplementary section labels. Reviewer wording below is transcribed from the retained original comments. The marked copy shows changes since the local scientific baseline, not a tracked comparison with the submitted paper.

Editor

Comment. In particular, the scientific gap and novelty of the study should be more clearly established and supported by a comprehensive and balanced review of the relevant literature. The rationale for the proposed workflow and its methodological choices should be strengthened and better positioned in relation to existing approaches. Further clarification and justification are also required regarding the interpretation and validation of the “hidden exposure” concept, the treatment of observability, the limitations of the GHSL data, and the selection of the InSAR datasets.

Response. We have restructured the Introduction around established InSAR quality control, published exposure practices and population areal interpolation, followed by three research questions. We also updated the literature comparison with recent work on nationwide population/building/infrastructure exposure (Xiong et al., 2026), building-level deformation-risk diagnosis (Yang et al., 2026), and uncertainty-aware coastal population-weighted relative sea-level rise (Oelsmann et al., 2026). These studies use broader inventories, target-level data or calibrated vertical-land-motion products; we distinguish their endpoints from our fixed-input allocation audit and do not imply that they validate our LOS observations. The revised gap does not depend on showing that earlier studies generally misused missing support. It asks how replacing a joint fine-scale population-support distribution with coarse summaries changes the reported quantity under fixed inputs, and under which conditions. The covariance decomposition explains the mech-

anism; the contribution is its quantitative evaluation across matched inputs, scales, origins and application settings, not a new mathematical identity.

The experimental response comprises matched-input scale/origin comparisons, controlled masking with known reference allocations, expanded sampling, alternative population products, a separate connected LiCSBAS experiment with excluded-edge residuals, and a Fresno application using actual published observation polygons. A further complete-interior analysis shows that the 1-km uniform-allocation difference remains 58.24-58.56% after excluding boundary-crossing cells. This addresses a possible geometric alternative explanation. Internal rechecks and relocated-input checks are documented in Supplementary S11.

The supported finding is that the population-weighted support summary changes substantially when the same population total is redistributed uniformly within coarse cells rather than retained at native model resolution. This allocation-and-aggregation effect is quantified with matched-input tests, controlled references and a published observation-footprint case. It is not a count of verified residents over subsiding ground. GHSL/WorldPop disagreement is treated as population-model sensitivity, and the incomplete Iran branch is excluded from quantitative applications.

An added Supplementary S15 diagnostic evaluates the population-support covariance identity and its Cauchy-Schwarz bound across square sizes, grid origins and two population surfaces. The numerical identity closes within 1.22×10^{-15} percentage points; we label this as an implementation check, not external validation or a confidence interval, and keep its pixel-centre sensitivity values separate from the direct-geometry primary estimates. The new crossed experiment tests the aggregation effect itself across population model and pair selection. With the common domain and population total fixed, uniform-minus-native differences are 55.38-55.64% and 51.29-51.51% for GHSL under 28 and 224 pairs, respectively; WorldPop gives 9.03-9.07% and 8.64-8.68%. All 16 model-sample-origin cases have a positive difference, while the magnitude is strongly model dependent. Four origins are sensitivity cases, not independent replicates. Pair number and composition change together. Locations: main manuscript p. 9, “Crossing population model and pair selection” paragraph; Supplementary S12 (Supplementary Information p. 13).

Changes and evidence. Main manuscript lines 47–94 (Introduction), 97–153 and 154–183 (estimand and allocation design), 339–380 (matched-input result), and 621–663 (method rationale and evidence limits). Supplementary S7–S15 and S18–S24 supply the additional processing, sampling, method and population checks.

Code archive and submission files

Response. The scientific reproduction package is publicly available at the fixed GitHub release v0.4.1, commit 5bf98f19abbed981989f9c97ef9b118eea524c5b. The reviewer guide gives installation, experiment-specific reproduction and numerical-comparison commands. The checkout bundles code, processed analysis inputs, reference tables and provenance; a companion asset supplies all six complete ACS files and the two official S20 Census archives with checksums. Other provider-controlled products and the complete raw processing stacks follow the documented acquisition routes. The earlier DOI 10.5281/zenodo.21444768 identifies v0.3.0 only; a revised DOI deposit remains pending. The main Code Availability statement and cover letter now identify the same public scientific release. The clean, numbered and local-baseline marked manuscripts carry the same scientific text.

Location. Main manuscript lines 695–704, Code Availability (clean manuscript p. 12); updated cover letter and versioned reviewer guide.

Reviewer 1

R1.1

Comment (original wording). The manuscript presents a valuable methodological contribution to the Earth science community by addressing an important but usually overlooked issue »> how

InSAR product support masks affect downstream exposure estimates. There are some critical point that must be addressed. 1- The "Hidden Exposure" claim is not validated. The authors should justify such a claim in a scientific way as they address other issues. The manuscript repeatedly emphasizes that it is not validating deformation in hidden cells, yet the central result (3.77 million people in "hidden exposure") is presented as if it carries decision-relevant weight.

Response. We agree that the submitted 3.772492-million result did not validate a count of people exposed to subsidence in unobserved locations. We withdraw that interpretation and its measurement-priority implications. The number sums whole coarse-cell populations selected by a center-majority rule; it cannot locate residents within unsupported subareas or establish subsidence there. We retain the value only for traceability and revise the title, estimand and conclusions accordingly.

The primary estimand is mean area-pair support deficit under a declared population model. Holding the geographic domain, support and population total fixed, native GHSL allocation gives 3.020 million allocation units, whereas uniform allocation within 1-km cells gives 4.763–4.770 million. These are model-allocation quantities averaged over area and pairs, not counts of particular residents. Conservation, direct geometry, the covariance identity and controlled-mask tests check the calculation against defined references; excluding boundary-crossing cells retains a 58.24–58.56% primary-case difference. These checks establish an allocation effect without claiming to validate the population model or missing deformation.

We now separately evaluate mean input-pair support and final-product quality on the same LiCSBAS grid and population field. Direct source-cell intersections across four origins give uniform-minus-native shifts of 3.066–3.189 percentage points for mean input support and 0.108–0.117 points for the final mask (Supplementary S13). The much smaller final-mask result is reported explicitly rather than using sampled support as a proxy for final failure. The separate processing experiment evaluates all 44 excluded edges, with pooled RMSE 0.984 mm, but shared dates, referencing and atmospheric signals limit independence. This is an internal reconstruction check, not independent GNSS/leveling validation (Supplementary S7).

For the actual final mask, unsupported pixels contain 0.2944 percentage points of total allocation. The worst-case finite-domain LOS-class intervals consequently have width 0.2944 pp at the -2, -5 and -10 mm yr⁻¹ cutoffs (Supplementary S21). This exposes the unresolved allocation without assigning motion to invalid pixels. Actual-LOS masking experiments use originally valid pixels with known hidden values; their sign-varying errors demonstrate dependence on the introduced missingness mechanism, without identifying motion in genuinely unsupported areas (Supplementary S14 and S22).

The Fresno examples test transfer to actual observation polygons and a separate published vertical-rate product (Supplementary S16–S17). They are not independent geodetic checks of our LOS field. Population benchmarks address a different question: GHSL/WorldPop and Census/CA-POP comparisons assess agreement and allocation sensitivity at the reporting-unit scale (Supplementary S19–S20). The ACS 2017–2021 comparison adds a coarser survey benchmark using the 80 published variance replicates: the aggregate Census total is within the ACS 90% MOE, whereas GHSL and WorldPop totals differ from the ACS point estimate by +10.57% and +6.03% (Supplementary S24). Neither these comparisons nor internal processing checks establish household locations or deformation in permanently unsupported pixels. The same evidence limits are now stated together in the main Discussion.

Changes and evidence. Main manuscript lines 131–153 (units and estimand), 488–524 (processing and observed-LOS checks), 573–582 (population benchmarks), and 655–663, 664–673 (evidence limits and withdrawal of the earlier interpretation). Supplementary S3–S4, S7, S13–S14, S16–S22 and S24 provide the detailed tables and scope limits.

R1.2

Comment (original wording). P 10: The classification of cells as "majority observable" vs. "not-majority-observable" simplifies a continuous problem. I think binary threshold is unsatisfying

the goals and scopes you defined before.

Response. We have made support continuous in the primary analysis and report population-, built-surface- and area-weighted distributions over $[0,1]$, rather than making the majority split the principal result. The mean deficit equals the integral of the weighted empirical support distribution; we verify this from the exact step distribution. Cutoffs 0.2, 0.3 and 0.4 and unwrapped validity alone remain sensitivity definitions.

The mean does not distinguish spatially persistent missingness from variable pass frequency, so these are not relabeled as the same object. Nor is the 28-pair diagnostic treated as a complete inversion network: it has 54 dates and 26 components. We report expanded 28/56/112/224-pair samples, temporal strata and 100 conditional draws per sample size, while keeping the separate connected processing experiment distinct. The original and nested 28-pair samples differ and are now explicitly identified.

Changes and evidence. Main manuscript lines 97–153 (continuous support and allocation summaries), 204–237 and 432–485 (sampling design and sensitivity); main Fig. 2 and Supplementary S2, S5 and S12.

R1.3

Comment (original wording). P9-10: "GHSL layers are used because they provide consistent public support for population and built-up surface across domains." »> GHSL data limitations are understated and the authors should improve this statements.

Response. Population-model sensitivity is now quantified directly. On the common valid domain the $q=0.3$ deficit is 13.37% for GHSL 2020 and 23.20% for WorldPop 2020. Equalizing the population totals still gives 2.989 versus 5.186 million units, so total-population differences alone do not explain the contrast. We report each model's domain, joint coverage, excluded population and normalized results.

GHSL 2015/2020 epochs are compared separately at fixed 30-arc-second resolution; native-model allocation is compared across reporting scales. Fractional WorldCover summaries preserve an explicit uniformity assumption within GHSL pixels. GHSL and WorldPop remain census/model-based allocations, not local demographic truth. GHSL population and built-surface layers share information, so a built-weighted comparator is not independent validation. Built surface is not occupied floor area; population is not vulnerability, loss or damage. Model differences are not calibrated error bars.

The new crossed experiment tests the aggregation effect itself across population model and pair selection. With the common domain and population total fixed, uniform-minus-native differences are 55.38-55.64% and 51.29-51.51% for GHSL under 28 and 224 pairs, respectively; WorldPop gives 9.03-9.07% and 8.64-8.68%. All 16 model-sample-origin cases have a positive difference, while the magnitude is strongly model dependent. Four origins are sensitivity cases, not independent replicates. Pair number and composition change together. Locations: main manuscript p. 9, "Crossing population model and pair selection" paragraph; Supplementary S12 (Supplementary Information p. 13).

Changes and evidence. Main manuscript lines 204–237, 432–485 and 573–582; Supplementary S6, S12, S19–S20 and S24. The combined evidence limits appear at lines 655–663.

R1.4

Comment (original wording). there are also some minor issues that must be double-checked by the authors. I recommend major revisions with particular attention to clarifying the weak points and the policy-relevant interpretation of the results too.

Response. We have checked the scientific quantities, units, dates, sample counts, figure captions, source attribution and cross-references. The revisions distinguish the original 3.772492-million center-majority total, an early revision estimand and the current 3.020-million mean deficit. They also distinguish source-product coverage, sampled pair support and final LiCSBAS quality.

The corrected Iran attribution, unconfirmed VLM scale, population-model dependence and temporal mismatches are explicitly retained.

We remove policy claims about a verified population hidden from subsidence assessment and do not propose a validated measurement-priority ranking. The practical recommendation is limited to transparent reporting of the domain, support definition, allocation assumptions, sensitivities and unresolved information. Internal numerical checks and PDF review support consistency; they do not add external physical evidence.

Changes and evidence. Main manuscript lines 638–685, Discussion and Conclusions (clean manuscript pp. 11–12); Supplementary S1–S2, S8 and S11 retain units, provenance, lineage and computational checks.

Reviewer 2

The manuscript "Auditing observability bias in InSAR-derived subsidence exposure estimates with a reproducible geospatial workflow" by Zixuan Liu and Wei Xiong discusses a very important topic regarding InSAR products and how they might translate into misleading exposure estimates. While the topic is relevant and the issues raised by the manuscript are valid, I have major concerns about the structure of the paper.

R2.1

Comment (original wording). 1- The problem formulation gives too little credit to established InSAR practices. It discusses the problem in the Introduction. For example:

"exposure accounting can convert this missingness into a denominator change."

"A processing team may publish a careful product with valid-support masks, while a downstream analyst may treat the remaining raster or point layer as a complete hazard domain."

"but they rarely audit whether non-observed radar support is being treated as absent"

I find the concern relevant and valid. But the literature review does not systematically show that previous InSAR exposure studies have failed to address areas with missing information. The manuscript claims that "When the resulting deformation map is intersected with population, built-up land or infrastructure layers, analysts often remove cells that are not valid in the radar product before exposure is summarized," without providing evidence from studies that follow this practice. In my opinion, this is the main weakness of the manuscript. If the issue is misuse of correctly documented InSAR products by non-specialist users, this should be stated explicitly and supported by multiple examples of studies where such problems occur, together with references to state-of-the-art studies that have tried to address the issue. In its current form, the manuscript sometimes reads as if the InSAR community has failed to recognize this limitation. However, it is well accepted in the community that InSAR observations are opportunistic by nature. In some cases, InSAR products are accompanied by quantitative information on the validity and quality of the observations. For example, LiCSBAS provides average coherence, the percentage of valid unwrapped pixels, etc. along with velocity and time-series products. The manuscript acknowledges that experienced InSAR users understand and correctly interpret areas where InSAR does not provide information. Therefore, the scientific gap should be demonstrated specifically at the point where InSAR products are used in downstream applications. Without enough evidence from previous studies, it is difficult to determine whether the paper addresses a genuine unresolved research problem.

Response. We agree that the original framing did not substantiate a general claim that previous studies mishandled missing support. We removed that claim and revised the Introduction to recognize established PS, SBAS, distributed-scatterer, LiCSBAS and MintPy practice and quality diagnostics. The literature review now describes the relevant exposure studies without labeling them as misuse examples. Our research question remains substantive without that prevalence claim: downstream summaries can change when fine-scale population-support structure is aggregated, even when the input support and population totals are unchanged.

Examples include Blackwell et al.’s stated uniform-within-community allocation and GNSS comparisons; Fernandez-Torres et al.’s coherence screening and GPS checks; Ao et al.’s urban population shares above rate thresholds; Ohenhen et al.’s census-block median-rate assignment and comparison with 154 GNSS stations; and Murray et al.’s explicit limitations for motion between buildings. We describe these as relevant practices, not demonstrations of misuse. Published Bangkok ground comparisons are summarized in the Introduction, with detailed station/time/support limitations moved to Supplementary S8.

The empirical question is: for a fixed population model, domain and support field, how much does spatial aggregation change the population-weighted support summary, under which conditions, and in which direction? Matched-input comparisons, complete-interior results, controlled sign reversals and model/sampling sensitivity answer it directly. This is a contribution about quantifying downstream aggregation effects, not an allegation that earlier authors erred. We distinguish LiCSAR sampled support from the WabInSAR source product’s validity mask; Fresno tests transfer to a published observation footprint with a different support definition.

The crossed experiment further demonstrates that this contribution is not confined to the primary GHSL calculation: its direction persists for WorldPop and expanded pair sampling, with a much smaller magnitude for WorldPop. We now explicitly restrict the approximately 58% value to the original primary case (main manuscript p. 9, “Crossing population model and pair selection” paragraph; Supplementary S12, Supplementary Information p. 13).

We have added a direct controlled benchmark of a published census-unit median-rate assignment rule against within-unit observed-population prevalence and centre sampling. Across 100 mask replicates, four fixed grid origins and four missingness mechanisms, the median-rate rule has about +1.9 pp mean signed error under random and clustered removal, while within-unit prevalence is +0.08 and +0.14 pp, respectively; under low-spread removal the median-rate rule is closer, and under high-spread removal all three underestimate. These are artificial-mask design sensitivities on originally valid LOS data, not evidence about genuine missing pixels (Supplementary S22, p. 24). This comparison both strengthens the downstream-method rationale and limits any claim of universal superiority.

Changes and evidence. Main manuscript lines 47–94 (reframed gap and literature), 381–431 (controlled estimator comparisons), and 621–663 (method scope). Supplementary S8, S10 and S22 provide the literature/method comparison and published-rule benchmark.

R2.2

Comment (original wording). 2- The manuscript explains how the proposed workflow works, but it does not adequately explain why this particular method is chosen. State-of-the-art methods and alternative approaches should be discussed and compared with the proposed method. The Materials and Methods section also contains very few references to previous methodological works. As a result, it is difficult to determine how the proposed approach relates to existing methods and why the specific choices made in the workflow are preferable or necessary.

Response. We now justify the workflow as a mass-preserving comparison of methods estimating the same declared quantity, and explain when each allocation rule is appropriate. The revised Methods and Supplementary S10 add established alternatives and methodological references, including the dasymetric comparison of Eicher and Brewer and the areal/ancillary allocation work of Mennis. We compare inputs, assumptions, interpretation and evaluation for center sampling, uniform areal allocation, native-model allocation, built weighting, observed-domain overlays, processing diagnostics and deformation imputation/additional sensors.

For methods estimating the same quantity, support, domain and population totals are fixed. Four scales and four origins test center and uniform allocation against the native model, while local errors and a covariance decomposition explain the discrepancy. The new complete-interior result remains 58.24-58.56%, so the primary effect is not explained by boundary-crossing cells. The controlled-mask experiments retain both signs of error, local cancellation and small aggregate-bias

cases. Fresno repeats the comparison using actual polygons; the small outside-support denominator is accompanied by an error normalized to the whole AOI population.

Native allocation is useful when retaining the selected model’s fine structure is appropriate; it does not validate that structure. Uniform allocation is adequate when the relevant covariance is negligible at the required scale. Built weights add assumptions and share information with GHSL. Imputation, sensor fusion and ground measurements address missing deformation under additional observations or assumptions; we do not claim numerical superiority over such different-target methods. The rationale for our protocol is conserving and exposing the consequences of known inputs, with quantified applicability conditions.

To test method sensitivity across inputs, the additional 2-by-2 model/pair comparison repeats the native-versus-uniform calculation under identical domains and totals. It preserves the population-support product before projection and reports signed differences against both the native deficit and the whole population. The 25-m integration diagnostic changes the result by at most 0.000436 percentage points of total population. The full 16-row table and numerical checks are supplied in Supplementary S12 (Supplementary Information p. 13).

We add one external population benchmark at the Fresno block-group scale. Against ACS 2017–2021 estimates, the 2020 Census P1 total differs by -0.36% (inside the aggregate ACS 90% MOE), while the GHSL and WorldPop 1-km totals differ by $+10.57\%$ and $+6.03\%$. The exact aggregate ACS uncertainty uses the 80 published variance replicates. This identifies model disagreement at the reporting-unit scale and does not establish which raster is correct at household scale (Supplementary S24, pp. 26–27).

We also report worst-case finite-domain bounds for LOS threshold-class population allocation under the actual full-stack mask. The unsupported allocation mass is 0.2944 percentage points of the fixed domain, giving class-share widths of 0.2944 pp at the -2 , -5 and -10 mm yr $^{-1}$ thresholds (Supplementary S21, p. 23). These bounds make the unresolved allocation explicit; they do not infer motion in unsupported pixels. A prespecified-tolerance interpretation of the scale/origin covariance bound is added in Supplementary S23, p. 25.

We added a paired LiCSBAS analysis that holds the population field, processing grid and coarse allocation constant while separately evaluating mean support across 437 inputs and the final full-stack quality mask. Direct intersections of source cells with the 1-km reporting grid verify the geometry independently of raster reprojection. The measured effects differ materially by support definition, demonstrating why alternative methods must be compared on the same estimand and why sampled input support should not be used as a stand-in for the final product mask. We report the four-origin sensitivity and full numerical audit in Supplementary S13.

We added a fine-resolution population-allocation sensitivity using official 2020 Census P1 block totals and the CA-POP 100-m dasymetric surface in Fresno. Normalizing both GHSL and CA-POP within the same blocks fixes their totals and isolates within-block allocation assumptions against the same DWR polygon-union support. The outside-support deficit is 2,016.3 Census allocation units for GHSL and 1,268.4 for CA-POP (a difference of 0.143 percentage points of the shared total); the direction varies across tracts. Because CA-POP is also built from these Census block counts, this is model sensitivity, not independent population validation. We quantified spatial dependence among the block-level unsupported fractions using binary queen-contiguity Moran’s I : the CA-POP-minus-GHSL difference field has $I = 0.399$ on 17,083 neighboring block pairs. This is reported descriptively for the fixed domain without a sampling inference. We also tested 250–2,000-m grids at four origins for center and uniform allocation, reporting local absolute errors, ranking overlap and a Cauchy–Schwarz bound; ranking overlap declines at coarser scales and center sampling varies with origin. The new tests support explicit scale/origin reporting rather than a universal correction factor (Supplementary S20, p. 22; main manuscript Fresno application).

The observed-LOS masking experiment also evaluates the proposed allocation choices against a known class-population reference created by masking pixels with measured LOS rates. Its results show good aggregate behavior for within-block prevalence under random masks but larger, sign-varying errors under spatially clustered or quality-selective masks; nearest-centre sampling can be

substantially biased even under random masks. This strengthens the comparison with an observed processing product while keeping the estimand explicit: only hidden values inside the originally valid domain are tested. The experiment does not supply evidence about permanent no-data areas.

Changes and evidence. Main manuscript lines 154–237 (matched-target choices and tests), 339–380 and 381–431 (allocation and controlled-mask results), 488–524, 573–582, and 621–663 (applicability). Supplementary S3–S4, S10–S15 and S20–S24 contain the full comparisons, bounds and sensitivity tables.

R2.3

Comment (original wording). 3- The "Iran public InSAR products" is based on relatively old study from 2008. More recent nationwide studies of subsidence in Iran are available, including Haghighi and Motagh, 2024 and Payne et al., 2025. InSAR datasets from both of these studies are publicly available. The manuscript should explain why an older dataset was selected instead of these newer products. I think the use of the 2008 dataset does not invalidate the workflow, but the it should be justified.

Response. We have removed the exploratory Iran branch from the quantitative applications and corrected its source attribution. We therefore no longer present the manuscript as having selected an older Iranian dataset for a completed case study, and we do not claim a new/old product comparison.

The checksum audit identifies the archived rate and mask as the 2014–2020 product accompanying Haghshenas Haghighi and Motagh (2024), rather than a 2008 nationwide rate product. The seasonal-amplitude archive does not match the companion checksum and is unused. Supplementary S1 and S8 distinguish that national annual-rate product from Payne et al.'s 2014–2022 selected-region vertical/east–west velocities and uncertainties. Those summary layers have different targets and are not pair-specific valid-unwrapped/coherence masks. Source interferograms remain separately available; we do not claim the newer products are inaccessible or inferior. The quantitative applications and conclusions are limited to the declared Chao Phraya and Fresno inputs.

Changes and evidence. Main manuscript lines 664–673, Earlier estimates and source attribution (clean manuscript p. 12); Supplementary S1 (p. 1) and S8 (p. 8).